Ehab Ashraf Mohamed

AI & Machine Learning Engineer

ehab44221@gmail.com | 01211114548 | Zagazig, Egypt 44511 | 20/11/2001

linkedin.com/ehab-ashraf | GitHub | Kaggle

PROFILE

Dedicated Machine Learning Engineer with a strong foundation in AI and Data Analysis. Adept at developing predictive models, analyzing financial trends, and building optimized portfolios for investors. Skilled in leveraging diverse datasets, including economic, financial, and astrological data, to generate robust and accurate predictions. Demonstrated ability to work independently and deliver data-driven insights for impactful decision-making. Known for attention to detail, adaptability, and strong problem-solving skills.

SKILLS

Programming & Languages
Python, SQL

Libraries & Frameworks
TensorFlow, scikit-learn, Keras, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning & AI
Supervised & Unsupervised Learning, Deep Learning, Transformers, NLP, Computer Vision, Time Series Analysis, Reinforcement Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning

MLOps & Tools
MLflow, Model Deployment, Model Monitoring, CI/CD, Git

Data Engineering & Tools
Web Scraping, Data Preprocessing, Data Mining, Jupyter, Data Visualization, Advanced Analytics

EXPERIENCE

Machine Learning Engineer & Data Analyst

Quantum Bits Company — March 2024 – Present

- Developed, deployed, and monitored stock price prediction models using financial, economic, and astrological data, improving prediction accuracy by **15%** over baseline models.
- Trained and maintained individual models for 50+ stocks, generating a 10-year track record of model performance to identify top-performing strategies.
- Analyzed quarterly financial peaks and global economic indicators, reducing forecasting error by 12% in quarterly revenue predictions.
- Built dynamic portfolios that outperformed the S&P 500 by 12% annually in 10-year backtests.
- Implemented MLOps practices (continuous training, deployment, and monitoring), cutting model update cycle time by 30%.
- Evaluated and compared model-based portfolios against benchmarks, achieving consistent alpha generation over multiple quarters.

Machine Learning Engineer

Freelancer, remotely 2022 – 2024

- Delivered ML solutions across NLP, computer vision, and predictive analytics, improving client model performance by up to 18%.
- Implemented regularization and hyperparameter tuning, boosting generalizability and reducing overfitting by 25%.
- Designed automated visualization dashboards, reducing client reporting time by 40%.
- Preprocessed and engineered features from datasets of 100k+ records, improving downstream model efficiency.

- Developed and deployed end-to-end ML models with monitoring pipelines, ensuring 99% uptime for client applications.
- Produced detailed performance reports with accuracy, precision, recall, and F1-scores, enabling data-driven decision-making.

PROJECTS

Portfolio Strategy Optimization (Quantum Bits)

Applied various investment strategies to the portfolio, evaluating their effectiveness on a quarterly basis. These strategies were used to optimize portfolio performance and make dynamic adjustments based on changing market conditions. The goal was to select the best strategy for each quarter, improving returns and minimizing risk.

Portfolio Stock Price Prediction & Analysis (Quantum Bits)

Developed predictive models for stock prices using financial, economic, and astrological data. Each stock was trained individually, generating a track record to identify the best-performing models. Over 10 years of historical data were analyzed to capture market and economic trends. The portfolio performance was compared against S&P 500, showing superior results. Additionally, quarterly peaks and bottoms were studied to understand the driving factors behind market movements.

English to Arabic Translation using Transformers

Implemented a neural machine translation model using transformer architecture to translate text from English to Arabic, leveraging advanced deep learning techniques to achieve high translation accuracy.

Customer Sentiment Classification through NLP and Machine Learning

Applied machine learning and natural language processing (NLP) techniques to analyze customer sentiment in reviews, improving businesses' capacity to gain thorough insights from customer feedback.

ACHIEVEMENTS

- Ranked 9th in the Artificial Intelligence Department.
- Ranked 10th in college.
- Won fourth place in the hackathon competition.

EDUCATION

Bachelor of Computers and Artificial Intelligence 08/2020 – 08/2024

Benha University

Benha

Graduated with a cumulative GPA of 3.77 (A). Ranked 9th in the Artificial Intelligence Department and 10th in college. Possess a strong grasp of computer science fundamentals and specialized expertise in artificial intelligence, encompassing areas such as Machine Learning, Deep Learning, NLP, Computer Vision, Robotics, Transformers, Advanced Data Analytics, and Data Mining, acquired during my studies in computers and artificial intelligence.

CERTIFICATES

AWS Academy
Data Engineering, AWS Academy
(May, 2024)

Hackathon
4th place in the accessibility and entertainment track at the recent hackathon for emerging technologies to empower people of determination
(February, 2024)

LANGUAGES

English ●●●●○ Arabic ●●●●●